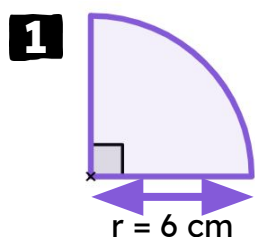




Area of composite shapes

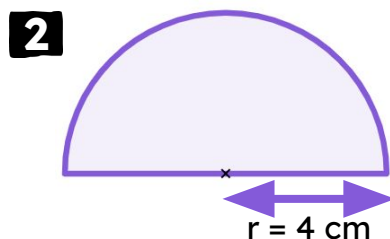
Task A: Finding the area of sectors of a circle

1) Match the circle sectors with the possible correct calculations and answers. Some calculations do not match with a sector.



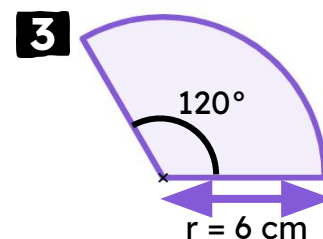
d $\frac{1}{4} \times \pi \times 6^2$

e 9π



c $\frac{1}{2} \times \pi \times 4^2$

f 8π



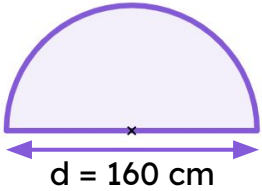
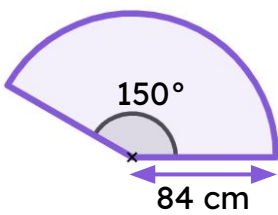
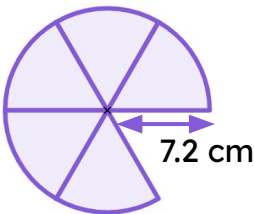
b $\frac{1}{3} \times \pi \times 6^2$

h 12π

2) Complete the table. Give your answer both as a decimal (rounded to 1 d.p.) and in terms of π .

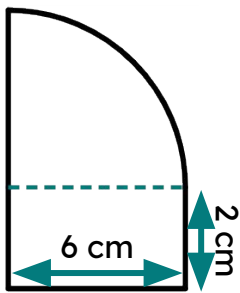
sector	step 1	step 2	answer (cm ²)
a)	$\frac{1}{2} \times \pi \times 30^2$	$\frac{1}{2} \times \pi \times 900$	450π 1413.7
b)	$\frac{3}{4} \times \pi \times 18^2$	$\frac{3}{4} \times \pi \times 324$	243π 763.4
c)	$\frac{2}{3} \times \pi \times 57^2$ or $\frac{240}{360} \times \pi \times 57^2$	$\frac{2}{3} \times \pi \times 3249$	2166π 6804.7

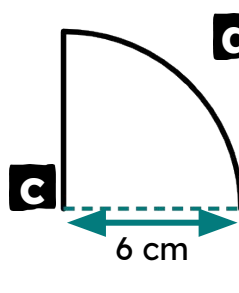
2) Complete the table. Give your answer both as a decimal (rounded to 1 d.p.) and in terms of π .

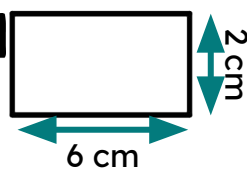
sector	step 1	step 2	answer (cm ²)
d) 	$\frac{1}{2} \times \pi \times 80^2$	$\frac{1}{2} \times \pi \times 6400$	3200π 10 053.1
e) 	$\frac{150}{360} \times \pi \times 84^2$	$\frac{150}{360} \times \pi \times 7056$	2940π 9236.3
f) 	$\frac{5}{6} \times \pi \times 7.2^2$	$\frac{5}{6} \times \pi \times 51.84$	43.2π 135.7

Task B: Finding the area of composite shapes with sectors

1) Match the composite shape to its component parts and calculations.

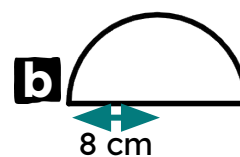
shape 1 

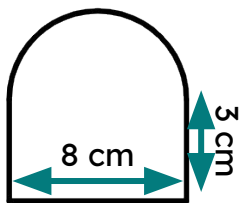
c 

d 

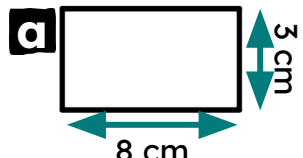
i $\frac{1}{4} \times \pi \times 6^2 + 12$

These did not match with either shape.

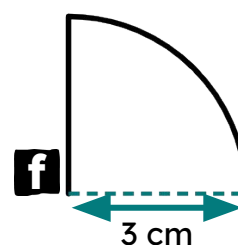


shape 2 

e 

a 

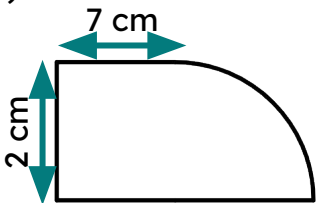
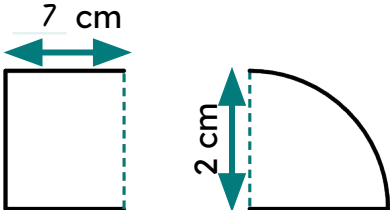
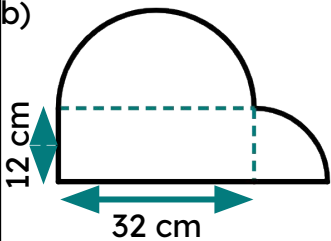
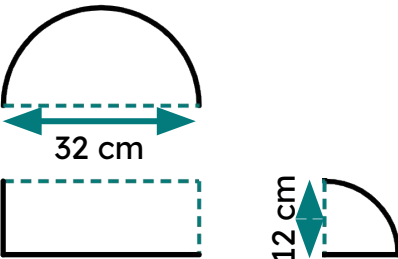
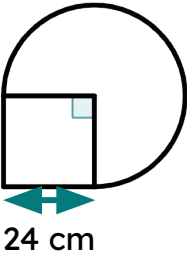
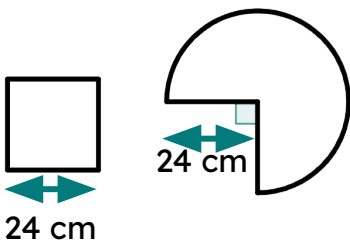
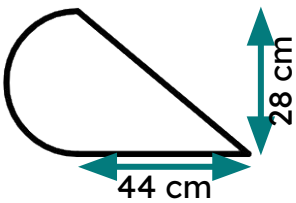
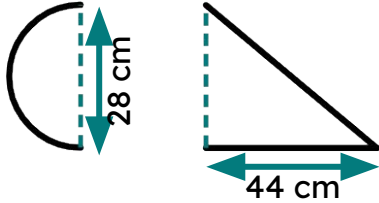
g $\frac{1}{2} \times \pi \times 4^2 + 24$



h $\frac{1}{2} \times \pi \times 8^2 + 24$

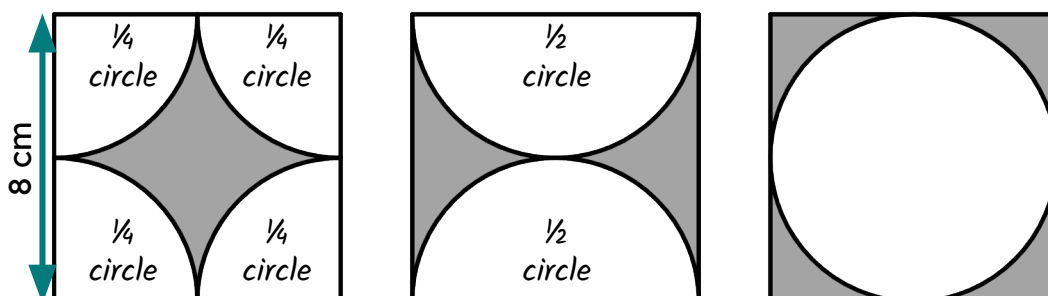
i $\frac{1}{2} \times \pi \times 3^2 + 12$

2) Complete the table. Give your answer as both a decimal and in terms of π

shape	shape broken into parts	calculations (addition)	answer
a) 		$2 \times 7 + \frac{1}{4} \times \pi \times 7^2$	$\pi + 14$ 17.1 cm^2
b) 		$\frac{1}{4} \times \pi \times 16^2 +$ $\frac{1}{4} \times \pi \times 16^2 +$ 12×32	$164\pi + 384$ 889.2 cm^2
c) 		$\frac{3}{4} \times \pi \times 24^2 + 24 \times 24$	$432\pi + 576$ 1933.2 cm^2
d) 		$\frac{1}{4} \times \pi \times 22^2 +$ $\frac{1}{2} \times 44 \times 28$	$98\pi + 616$ 923.9 cm^2

3) What do you notice about the areas of each shaded region?

Show your working to support your observations.



The area of each shaded region is equal, as the empty space in each shape can be rearranged to create one full circle. The area of the shaded region is $64 - 16\pi$.

